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RESEARCH ARTICLE

Identifying Irish Traditional Music Genres Using Latent Audio Representations

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ABSTRACT Irish traditional music contains a variety of features that help to understand the cultural idiosyncrasies of Irish people. One of the most prominent aspects is genre, which provides information on the musical structure, the composition's essence and, in turn, offers insights into the cultural nuances and behaviors inherent to a particular community. Identifying genre in traditional music can be challenging without prior specialized musical knowledge. Hence, the integration of technological tools becomes valuable, offering assistance in discerning and categorizing genres, thereby facilitating a broader appreciation and understanding of traditional musical heritage. In this study, we introduce a novel system designed to classify the genre of Irish music by analyzing a latent representation of the raw audio extracted from synthesized scores. We evaluate prevailing genre classification models for Irish traditional music and highlight the need for specialized approaches to capture its unique characteristics. We demonstrate how this novel latent representation could be effective in the task of classifying Irish traditional music genres. To support this research, we compile an extensive score dataset of different types of Irish traditional music scores from The Session platform and the Irish Traditional Music Archive (ITMA). This new dataset can aid in the research and exploration of Irish music genres. We have also developed a practical tool using XML and ABC data formats to assist the ITMA in efficiently cataloguing newly acquired data.

INDEX TERMS Music genre classification, music scores dataset, Irish traditional music, latent audio representations, raw audio, jukemir, MIR, musicxml, abc notation.

I. INTRODUCTION

The preservation of music as a cultural heritage of a country is receiving a great deal of attention, especially within the European Union [6], where the significance of traditional music has been highlighted through specific funding and calls. In particular, the richness of Irish traditional music, which has established itself as one of the country's most important symbols and emblems [37], has been a central focus of research and action within the realm of popular music. Irish traditional music holds a deep historical and

cultural wealth, passed down from generation to generation through the oral tradition [36]. However, in the era of globalization and rapid cultural diffusion, this rich musical heritage faces significant challenges.

One of the major challenges for Irish traditional music is the risk of loss due to the oral nature of its transmission. Over the centuries, this music has been passed down from master to apprentice and is often transmitted without written notation [36]. However, in the present day, with the evolution of society and communication, oral transmission runs the risk of becoming less prevalent, leading to a decline in the authenticity and preservation of traditional music.

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Furthermore, globalization has also led to cultural homogenization in many aspects, resulting in the loss of the unique cultural identity that traditional music provides to a region or nation. As commercial music and global music trends dominate the scene, authentic musical expressions deeply rooted in a country's history may be marginalized, endangering the rich musical diversity that defines local communities.

Therefore, the focus on preserving and digitizing Irish traditional music has gained relevance. The digitalization of this musical heritage allows for its conservation and accessibility on a global scale, not only preserving Ireland's cultural richness but also promoting greater understanding and appreciation for the musical tradition among international audiences. The creation of digital archives and the availability of traditional music online ensure that future generations can easily access these musical gems and learn from them. In line with this idea, the Irish Traditional Music Archive (ITMA) was established in 1987, aiming to acquire, preserve, organize, and promote Ireland's musical and cultural heritage [4].

Moreover, efforts to preserve traditional music also have an impact on cultural tourism, as authentic music and local traditions attract visitors interested in immersing themselves in a region's culture and history [21]. This not only enriches the local economy but also fosters a sense of pride and appreciation for cultural heritage.

Governments and cultural organizations within the European Union have been actively working to support and fund projects dedicated to preserving traditional music, including Irish traditional music. These efforts include promoting festivals, concerts, and workshops related to traditional music [5], [7], as well as implementing educational programs to transmit this knowledge to the younger generations and conducting research projects to expand the current knowledge about traditional music [1], [2].

Given this information, it is clear that traditional music archives, such as the ITMA, have the challenging task of digitizing all the materials they receive, which are often found in various formats, such as audio, photos, videos, posters, books, and more. Among these formats, audio archives may be of particular interest for collecting popular traditional music. However, working with such archives can be quite tedious and involves considerable effort for documentalists. To make these tasks easier, archives can rely on advances in Artificial Intelligence (AI). In particular, AI models can be trained with existing archive materials and other data sources to process and streamline documentation.

Furthermore, many archives also lack tools that allow users to explore their collections more interactively and visually, making traditional music more engaging. These tools help expand traditional music and promote awareness of such initiatives and archives, encouraging other citizens to donate or make their private collections or materials related to traditional music available to the archive. Therefore, AI-based solutions are not only useful for documentalists but

also for citizens, as they can explore archive collections or even better understand the intricacies of traditional music.

Drawing on this, our article aims to contribute to the generation of AI-based tools that can be beneficial to the community. Therefore, we present the following contributions: (I) the creation of a new genre/subgenre classification model within Irish traditional popular music based on latent representations of the music raw audio; for this, we will need (II) to compile and create a new dataset of Irish traditional music, and finally, (III) to provide a tool that allows for classifying audio and/or files in various formats into the different genres/subgenres used during the study and creation of contributions (I) and (II).

The rest of the article is structured as follows: section II explains previous research conducted in the area of Irish traditional music as well as the peculiarities of this type of music; section III presents the proposed system for generating the developed AI model; section IV showcases the case study, including an analysis of the dataset used; section V presents the results obtained by the AI model and a discussion thereof; and last but not least, section VI provides the conclusions reached in the study and explores future lines of research.

II. BACKGROUND

A. MUSIC AND GENRE CLASSIFICATION

Music classification [38] is a task in the field of Music Information Retrieval (MIR), which aims to computationally capture the semantic aspects of musical compositions. Its main purpose is to extract the relevant musical features inherent in a given piece of music. This general task encompasses a wide range of specific classification tasks, covering various facets such as genres, moods, and instruments, as well as more overarching notions such as musical similarity and auditory tendencies. The data acquired in this task can be effectively used in various applications, including but not limited to music recommender systems, playlist construction and semantic search functionalities.

The so-called genre classification can be found among the previous tasks, and labeling a tune with a genre is a common task that an archive must carry out when new material arrives. In the literature it is possible to find two ways of formulating the task of genre classification:

- **Single label classification:** where a flat and mutually exclusive genre taxonomy is defined, i.e. genres such as blues, pop, rock, etc., and songs can only belong to a single genre. This is the simplest approach in the literature, with landmark datasets such as GTZAN [33].
- **Multi-label classification:** This approach is already more correct and would be advocated by many musicologists. In this approach, a song can have more than one genre label. In general, this seems to be the correct approach. Recent work in the literature already approaches the problem in this way, and many datasets [16] already provide multiple tags for each song.

This approach would be more appropriate for a music tagging problem [25].

B. GENRES IN THE TRADITIONAL MUSIC OF IRELAND

Throughout this section, some of the interesting points from the Ethnomusicology of Irish traditional music are briefly explained. In particular, we will introduce the classes that are to be inferred by classifiers in this work.

It will be assumed in this paper that as long as a tune is part of the Irish repertoire of Irish traditional musicians and share Irish features, then it will be considered part of Irish traditional music, regardless of the fact that it may have originated in other regions or cultures. Therefore, the repertoire of Irish traditional musicians also assumes that other folk songs and tunes, such as Polish and English songs, can also be considered part of the Irish tradition.

1) POLKA

This musical genre originated in Bohemia during the 18th century and arrived in Ireland by the late 19th century. Primarily, this genre developed in the southwestern regions of Ireland (Cork, Kerry, and Limerick), although it gradually spread throughout the entire territory of Ireland. The polka is characterized as a popular dance in 2/4 time and is commonly performed with marching rhythms [23].

2) STRATHSPEY

This type of tune is original to the region of Scotland and emerged in the mid-18th century. It wasn't until the late 19th century that they arrived in Donegal (Ireland), although in Ireland, strathspeys were never used as dance tunes. These tunes are noted in 4/4 time and are characterized by the emphasis on each beat of the bar and by the "Scots snap", a type of dotted rhythm where the long note is preceded by a short note [23].

3) JIG

Jig tunes are used in competitive or exhibition dance settings by both men and women. The tempo used is 6/8, and the commonly used pattern is two groups of three quavers each. Additionally, the melodies are structured in two main sections of eight bars each; each of these sections is repeated once before moving on to the next section [23].

4) SLIP JIG

This distinctive type of jig differs from the previous one as it uses a 9/8 time signature, resulting in a rhythmic pattern of three groups of three quavers each. Unlike the previous jig types, its two parts of eight bars each are not repeated, and it is designed to be danced in a group setting. Popularly known as "the queen of step dances", it earned this title due to the graceful dance style performed when it is played [23].

5) REEL

Reels, among the most ancient Irish dances [34], are executed in 4/4 time signature and are believed to have debuted in Ireland during the late 1700s. This lively dance form can be performed individually or in groups and finds expression in various settings, including competitions, exhibitions, and social gatherings. A distinctive feature of the reel lies in its composition of two groups, each consisting of four notes, resulting in an eighth-note bar. Within each group, the presence of two heavy-light pairs creates its characteristic rhythmic pattern.

6) MARCH

According to [34], the march, originating from military activities, evolved into dance in Ireland to accommodate the abundance of tunes available and to introduce rhythmic diversity in céilí dancing. Its tempo served functional purposes, such as facilitating efficient travel, marking victory in celebration, or evoking somber tones to represent death or defeat. Military marches were categorized as "slow", "parade", and "quick", the latter of which was integrated into early 20th century popular dance as the "quickstep". While some old Irish examples adopted different time signatures like 9/8 and 3/4, the most common time signatures in the Irish tradition were 4/4 and 2/4, with 6/8 also being popular.

7) HORNPIPE

The hornpipe, set in 4/4 time, is defined by its distinct dotted rhythm and emphasis on beats one and three. Its introduction to Ireland originated from England during the late 18th century, often presented by professional dancers during intermissions of theatrical performances, and carries historical associations with maritime themes. Consequently, Irish hornpipes acquired an exhibitionistic character. It is noteworthy that while some hornpipe compositions may not be notated with dotted rhythms, they are customarily performed with a swinging feel, even when the sheet music lacks dotting.

8) THREE-TWO

Three-twos represent a distinct march-like musical composition, executed in a 3/2 time signature, which can be traced back to its potential roots in the adaptation of the triple meter to the hornpipe genre.

9) BARNDANCE

The barndance, a traditional genre primarily set to a 4/4 rhythm, adopts a unique dance style in North County Antrim, where it is performed to the lively 6/8 time signature, reminiscent of marching practices. Historically, barndances thrived as a favored social dance until the 1950s. The name "barndance" derives from the customary practice of dancing in large sheds known as "barns", which was prevalent before dedicated social and meeting halls became available. This

association with barns imparts a distinct rural character to this genre [34].

10) WALTZ

The waltz, a dance form in 3/4 time, is notably characterized by the prominent emphasis placed on the first beat in each bar. Although its precise origins remain uncertain, it is believed to have emerged as early as the 14th century, independently from the European minuet tradition. Typically rendered through singing or played on the fiddle, waltz tunes often carry agricultural associations [34].

11) MAZURKA

A dance-form performed in 3/4 time, the mazurka gained significant popularity among Irish musicians, particularly in Donegal, where it became prevalent in the first half of the nineteenth century. Originating in the Polish province of Mazovia during the 1500s [34], the mazurka stands apart from its relative, the waltz, by emphasizing the second beat instead of the anticipated first beat among its three beats. While the rhythmic structure of Donegal mazurkas adheres entirely to the definitive Polish pattern, the phrasing employed aligns with other Irish rhythms, showing no connection with the mazurkas of Eastern Europe.

12) SLIDE

Slides, a melodic type closely related to jigs, are distinguished by their quicker tempo and adherence to a 12/8 time signature. The prevalent rhythm centers around the interplay of crotchets and quavers, engendering a distinctive sensation of long and short notes. Primarily intended for dance, slides intricately synchronize their long-short rhythmic patterns with the graceful movements of the dancers, creating a seamless union between music and motion.

13) AIR

It corresponds to the abbreviation for “slow air”, which is a musical composition found in diverse meters, yet generally lacking strict adherence to any specific meter. The slow air assumes the form of an “open-ended” melodic structure. Frequently, a slow air consists simply of the melody from an existing song, such as “Dear Irish Boy”, “May Morning Dew”, “Wounded Huzzar”, or “Anach Chuain”. However, a significant number of classic pieces, including “Lament for Staker Wallace”, “O’Donnell’s Lament”, and “Gol namBan san Ár”, exist independently without any direct association with an original song.

14) PLANXTY

This is a type of tune created as a tribute to a patron, primarily linked to the seventeenth-century harper Turlough Carolan. The origin of the term might stem from the Irish word “sláinte”, signifying “health”. Within Carolan’s repertoire, a few planxties are found in 6/8 time, with “Planxty O’Rourke” also existing in a variant time signature of 3/4.

15) HIGHLAND FLING

The highland fling stands as the national dance of Scotland, characterized by its spirited nature demanding a graceful equilibrium and precise execution. Believed to have originated as a triumphal dance for a solitary male performer, the highland fling was historically enacted after battle. Executed in a 4/4 time signature, this dance entails a sequence of intricate steps performed in a stationary position.

16) FLING

Fling pertains to a dance and dotted-rhythm tune category associated with the hornpipe, its nomenclature drawing influence from the Scottish “highland fling”. These flings gained popularity in Ireland during the latter part of the nineteenth century and continued to be performed by touring musicians, exemplified by artists like Jimmy Shand, throughout the twentieth century. Typically showcased during “céilí and old-time” dance gatherings, the collection curated by Roche comprises six tunes in this style, encompassing both Scottish-origin melodies and numerous 4/4 compositions. By altering accentuation, Roche’s collection can effectively evoke the rhythm characteristic of this genre.

17) CLOG

The tune aimed to the “clog dance” which is a unique form of dance in which the dancer emphasizes the rhythmic patterns of their feet through the use of wooden-soled shoes, known as clogs. This distinctive style of dance can be observed in various dance forms, such as certain bourrées of Auvergne, Swiss Ländler, and frequently in Irish step dances, including solo jigs, reels, and hornpipes.

C. PREVIOUS WORKS ON IRISH GENRE CLASSIFICATION

Before looking at specific works on the classification of genres in Irish traditional music, some previous works on the classification of musical genres, in general, will be briefly discussed.

The classification of music in terms of musical genre has been analyzed extensively in the existing literature. However, the techniques used have been changing over the years as explained by Ndou et al. [26] in their work; the AI algorithms used for genre classification can range from traditional Machine Learning (ML) algorithms [20], [26], such as decision tree-based algorithms, rule-based algorithms, near-instance-based algorithms, etc., to algorithms based on neural networks or Deep Learning (DL) [31], [40], such as Convolutional Neural Network (CNN), Recurrent Neural Networks (RNN), etc. These latter algorithms have demonstrated their capability of obtaining better results than traditional ML algorithms, although they are generally more expensive in terms of computational cost.

On the other hand, authors in the literature have used inputs encoded in different ways to obtain the musical genre. Works such as [27] and [29] use musical features extracted from audio and/or melody metadata. Other authors, on the

contrary, use an approach focused on image classification and computer vision, obtaining the spectrogram of the audio and using this image as input for the DL algorithm [10], [17]. A very recent option, which has already been used by some authors [14], is to use the audio and large DL models based on the Transformers architecture [35].

Analyzing the literature focused in Irish tradition, we can highlight the work carried out at the Dublin Institute of Technology [9] where they have developed a model for inferring the rhythm of traditional Irish folk tunes. This is not the only work they have done, they have also carried out a small classification between five possible genres of the songs present in the dataset used in the study. To solve this task, the authors obtain a vector of characteristics that they extract from the spectrogram of the audio.

Another interesting work in the field of Irish traditional music is the one by Shingte and d'Aquin [30]. In this case, the authors also aim to make a grouping of the existing sub-genres within Irish traditional music to see the similarities between them and to look for connections or ramifications between the various music sub-genres. For this purpose, they use a type of ML algorithm called unsupervised learning which is provided with features extracted from both audio files and structured files with musical information (MIDI files). This system provides a tool that allows users to observe the most similar songs within the dataset to one provided by them. However, this approach does not use a technique that allows for automatic classification within the sub-genres of Irish traditional music.

Another interesting study is the one developed by Martins and Silla Jr. [23]. In this case, they use high-level symbolic features extracted from MIDI files to classify Irish musical subgenres using decision trees. The results obtained by this study are quite good for some of the subgenres, but in others, they are quite poor. As can be seen, a classical ML algorithm is used in this study and the results can likely be improved with more recent solutions.

In contrast, other studies deal with traditional music from a more international point of view, focusing their study on the analysis of traditional music from several European countries, including Irish music [15]. This work uses hidden Markov models to define a model that classifies music according to its country using musical features extracted from the audio. Like other works described above, it uses classical models rather than more recent ones that might allow better results in terms of the performance of the model.

Hence, it can be seen that the literature has not yet explored the new latent representations of the latest deep learning approaches, such as Jukemir [11], and whether they can provide a breakthrough in performance within this task.

D. EXISTING TOOLS ABOUT TRADITIONAL IRISH MUSIC

Traditional Irish music holds significant cultural value and has garnered widespread interest among enthusiasts and researchers alike. As computational technologies advance,

various tools have emerged to aid in the identification, exploration, and preservation of this rich musical tradition. In this subsection, we highlight some of the existing tools for the following purposes:

- **Tunes Identification and search:** Accurate identification of traditional Irish music tunes is crucial for cataloging and organizing this vast musical repertoire. One notable tool for tunes identification is FolkFriend [3]. FolkFriend provides a user-friendly platform for users to search and identify specific traditional Irish music tunes. TunePal [19] is another valuable tool for traditional Irish music enthusiasts. It provides a social platform for musicians to search tunes, scores, compositions, and interpretations of traditional Irish tunes.
- **Tunes features exploration and geolocalization:** Geolocalization plays a vital role in understanding the distribution and regional variations of traditional Irish music. An interactive map tool [8] allows users to visualize the geographic spread of different traditional Irish music tunes and subgenres. Users can explore tunes across different regions, gaining insights into regional diversity and cultural influences.
- **Tunes Generation:** Tunes generation is an exciting area of research that explores the computational creation of traditional Irish music. Recent advances in deep learning have paved the way for innovative approaches to generate new tunes that adhere to the style and conventions of traditional Irish music. Works such as [32] and [39] have proposed methods that leverage machine learning techniques to produce novel traditional Irish music compositions.

The advances in the design of these kinds of tools around traditional Irish music significantly help to preserve, explore, and promote this musical heritage. Whether identifying tunes, mapping their geographical distribution, generating new compositions, or facilitating musical collaborations, these tools provide valuable resources for researchers, musicians, and ethnomusicologists alike. As technology continues to evolve, we are contributing with a new tool for classifying the newly collected tunes.

III. PROPOSED CLASSIFICATION SYSTEM

In this paper, we propose a system capable of classifying the Irish musical genre of a song from a latent representation of the raw audio obtained from a synthesized score. It is proposed that the input data to the system is sheet music in MusicXML or ABC notation, transformed into MIDI format, and then synthesized into raw audio used to obtain the latent representation (also referred to as “embedding” in the literature).

A diagram showing the general architecture of the proposed system is shown in Figure 1 below.

The embeddings are obtained using a pre-trained machine learning model. These models encompass deep learning architectures where a specific N-dimensional vector is

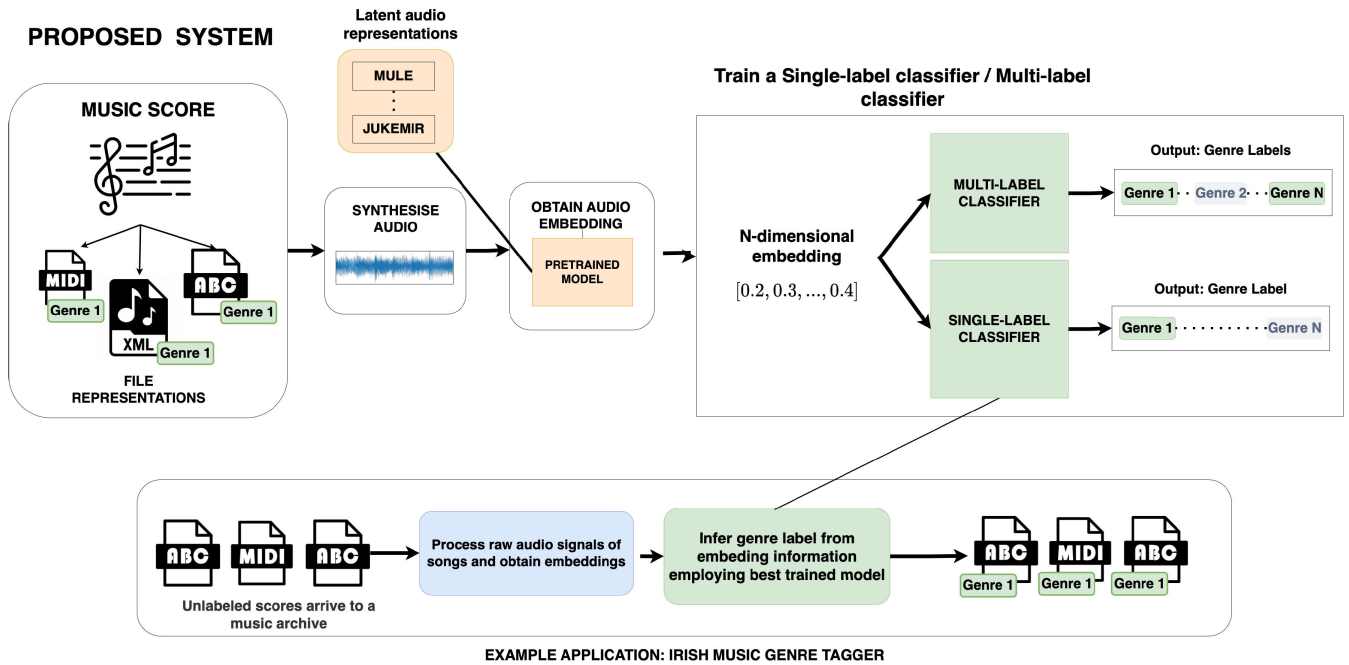


FIGURE 1. A diagram of the proposed system architecture for music genre classification.

extracted from one of the intermediate layers of the neural network.

This embedding is considered as a feature vector and used as input to a machine learning model. Subsequently, these representations are trained on various state-of-the-art classifiers and their performance is evaluated in a single-label classification task setting. This schema has been chosen due to the nature of the data collected, as they have disjoint generics, however, the same approach can be used in a multi-label setting, as has been shown in previous works [25]. Finally, the best-performing classifier is the one used in the system for tagging new songs. This system can be used as an auto-tagger in the ITMA archive for new collections of scores in the future.

The following section shows the latent representations selected for this work and discusses those chosen for the proposed system.

A. SONG REPRESENTATION

Different approaches have been followed in the scientific literature to obtain a latent representation of a score and classify its genre. Some authors have chosen to use MIDI or ABC encoding directly while others have chosen to obtain a raw audio first from the symbolic representation and then get a latent representation of the raw audio, as proposed in this work.

Within the approaches that use raw audio as input, we highlight representations such as constant-Q chromagrams (CHROMA) [17], Mel-frequency cepstral coefficients (MFCC) [10], Choi et al. approach (CHOI) [14], MUSICNN or CLMR [26].

In the last few years, two approaches have been reporting positive results in the genre classification, namely JUKEMIR and MULE. These representations have been chosen to evaluate their effectiveness in the proposed system of this work because they have proven their success in classification tasks. Furthermore, they have been postulated as the current leaders in the task in Music Information Retrieval of genre classification, according to the MARBLE¹ benchmark that systematically evaluates works in MIR.

B. JUKEMIR

One of the leading approaches in the current state of the art is presented by Castellon et al. [11]. Their work introduces the utilization of the internals of the Jukebox system [18], a system dedicated to generating music. Jukebox leverages the latest advancements in Natural Language Processing (NLP) and employs a technique where raw audio signals are transformed into discrete sequences and are directly used as input for NLP models, specifically the Transformers. Through this approach, Jukebox trains a Transformer language model, which operates as an autoregressive generative model, using encoded audio data from a massive dataset of 1 million songs. This training process is referred to as Codified Audio Language Modeling (CALM).

The previously mentioned model has demonstrated its effectiveness in music generation. The study by Castellon et al. [11] illustrates the feasibility of deriving valuable representations from this model by leveraging the activations of layer 36. This process results in the creation of a

¹<https://marble-bm.shef.ac.uk/leaderboard>

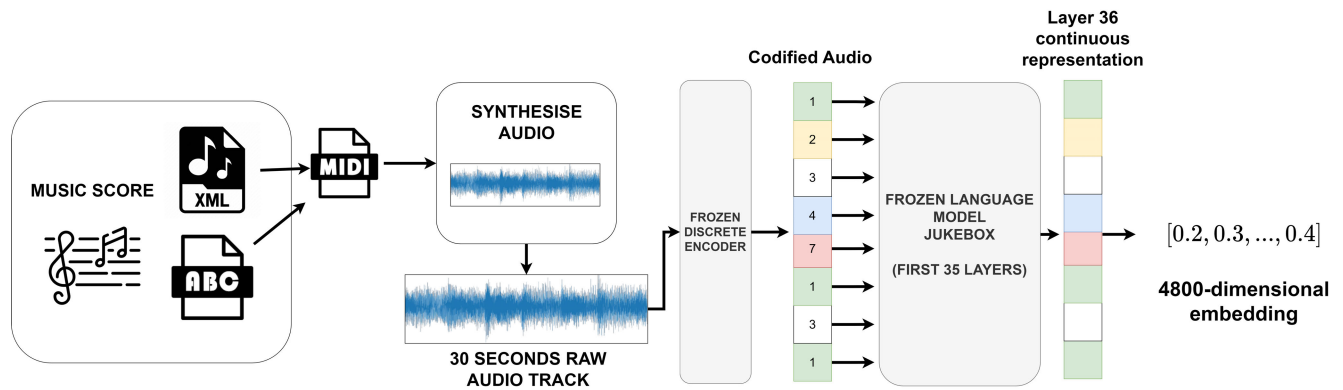


FIGURE 2. A diagram of the Jukemir's embedding extraction procedure.

4800-dimensional embedding that serves as a representation of the song.

A diagram showing the embedding extraction procedure of Jukemir is shown in Figure 2.

C. MULE

The Mule approach is a recent work in the literature proposed by McCallum et al. [24]. It is characterized by proposing different types of embeddings, using supervised and unsupervised approaches to train the pre-trained model. For supervised learning, they train models on large datasets of annotated magnitude log-mel spectrograms, both in the music domain and in the general audio domain. For unsupervised learning, they train contrastive models using SimCLR loss [12], [13] on the same sets of magnitude log-mel spectrograms. It uses SimCLR concepts, presents results in SOTA on several downstream tasks, and also obtains a more compact representation of a 1728-dimensional embedding, and requires fewer computational resources.

D. MERT

The Acoustic Music Understanding Model with Large-Scale Self-supervised Training (MERT) operates as a framework designed to unravel acoustic patterns inherent in musical audio signals [22]. MERT leverages an extensive corpus of unlabeled audio data to autonomously construct a representation of musical features.

MERT² harnesses a deep neural network architecture that includes an initial phase of unsupervised pretraining to discern salient patterns within the raw audio spectrograms. In this phase, MERT refines its internal representations, progressively encoding an encompassing understanding of musical characteristics.

Following the unsupervised pretraining, MERT embarks on a phase of supervised fine-tuning. A curated dataset of labeled musical examples is introduced, allowing the model to align its learned representations with specific musical attributes, such as instrument classification, genre

identification, or chord progression analysis. This convergence of unsupervised and supervised learning allows MERT to map raw audio spectrograms to a high-dimensional latent space, wherein musical information is compactly encoded. The final layer of the neural network outputs an N-dimensional vector representation that encapsulates the essence of the input audio. This vector serves as an embedding that represents the original audio waveform.

IV. CASE STUDY

In this section, we will explain some aspects that we consider important to explain how the research described in the article was carried out. These aspects not only illustrate the following steps but also help other researchers to understand some peculiarities and allow them to reproduce or adapt the research study when necessary.

A. DATASET

The dataset used to carry out the present research is a novel dataset that focuses on songs from traditional Irish music. As this is a completely new dataset we consider it essential to define how we have compiled and treated each of the elements of the dataset as well as the data sources from which they have been extracted. In addition, we will explain the structure and content of the resulting dataset.

1) DATASET COLLECTION

The dataset focuses exclusively on Irish traditional music, which may be one of the most widespread folk music traditions in the world. Indeed, this is because the Irish community has experienced migration movements over the years due to various economic, political, and social crises throughout the world and, as a result, they have spread their musical tradition throughout different territories. This is why Irish traditional music has a great variety of music genres/subgenres and tunes; most of them can be found in various archives or repositories.

Thus, to conform the dataset, two main data sources have been used. The following is a brief explanation of each of them as well as the records and characteristics retrieved:

²<https://github.com/yizhilll/MERT>

TABLE 1. Files genre occurrences in ITMA, The Session, and our final dataset.

Genre/Type	ITMA Occ.	TheSession Occ.	Final Dataset
Unknown	731	-	-
Reel	444	16275	16719
Jig	308	11521	11829
Hornpipe	288	3173	3461
Air	138	-	138
Strathspey	56	1161	1217
Clog	42	-	42
Highland fling	13	-	18
Fling	5	-	-
Polka	4	3650	3654
Planxty	3	-	-
Barn dance	2	1554	1556
Slip jig	1	1607	1608
March	1	1261	1262
Waltz	-	3298	3298
Slide	-	969	969
Mazurka	-	498	498
Three-two	-	358	358

- **ITMA:** as explained above, the ITMA is a music archive focused exclusively on Ireland's rich traditional music and provides a vast amount of information for users to consult. However, much of this information, such as audio files or digitized scores, is protected by copyright and therefore cannot be used in our research.

ITMA has provided us with a total of 2036 files in MusicXML format. These files correspond to Ryan's Mammoth collection, Bunting's collection, and Goodman's collection. Among the files provided, not all have a music genre/subgenre label; only 1305 of the 2036 files have a genre label. Table 1 shows the genre distribution of the files provided by the ITMA.

As shown in Table 1 in the second column the ITMA dataset is highly unbalanced and there are some musical genres that in our opinion overlap or cannot be related to strictly musical features. Therefore, we have decided to combine two genres, "Highland fling" and "Fling" into one single genre, "Highland fling". On the other hand, we have decided to remove the class "Planxty" as it is a genre defined strictly by the lyrics of the tune and not by musical features. Hence, out of the total number of files provided by ITMA, a total of 1302 are left.

- **TheSession.org:** as already mentioned, TheSession is a platform that focuses exclusively on traditional Irish music. TheSession has a vast number of files in different formats. Within our research we will focus our attention on the music scores available on the platform; in our case, we have been able to download a total of 45325 scores in ABC format or notation. These scores are accompanied by the musical genre of the songs as well as other information. In the third column of Table 1 we show the distribution by genre of the data obtained from TheSession.

After collecting these files together with their musical genre/sub-genre, either in MusicXML format (ITMA data) or in ABC format (TheSession) it is required to transform

these files into MIDI format (file extension *.mid) to later synthesize the audios of each of the files in WAV format. As a result, we will obtain synthesized audio from each score.

Moreover, we have also processed the MusicXML and ABC files to generate their alternative versions in ABC and MusicXML respectively. Consequently, we can provide other researchers with the entire dataset in both formats.

Afterward, every WAV audio file is used to extract its latent space representations, or embeddings, with the AI models explained in III. For this purpose, we use only the first 30 seconds of each of the synthesized WAV files.

2) DATASET DESCRIPTION

The resulting dataset is formed by a main file in CSV format in which the features of all the files are displayed. These features are the following:

- **Name:** it contains the name or title of the tune.
- **Source:** determines the source from which the tune was extracted, ITMA or TheSession.
- **URL:** web link to the tune where you can see more information about it.
- **Genre:** musical genre of the song.
- **Filename:** the name of the file; this name is used in each of the formats in which the dataset is provided (ABC, MusicXML, MIDI, and WAV).
- **Embedding_jukemir:** contains the embedding generated by the Jukemir model of the first 30 seconds of the tune it is an array of 4800 dimensions.
- **Embedding_mule:** contains the embedding generated by the MULE model for the first 30 seconds of the tune; it is an array of 1728 dimensions.
- **Embedding_mert:** contains the embedding generated by the MERT model for the first 30 seconds of the tune; this is an array of 1024 dimensions.

Likewise, the resulting dataset has the distribution of genres shown in the last column of Table 1. As can be seen, this is a highly unbalanced dataset, although this is a common problem in music collection works.

B. MACHINE LEARNING ALGORITHMS

An essential aspect of our system is the determination of the musical genre/subgenre of the tunes. For this purpose, the different embeddings of the tunes will be processed by an ML algorithm to obtain the musical genre/subgenre. To obtain the best possible genre classification model, three similar experiments have been designed. The difference between these experiments lies in the data that will be provided to the ML models, i.e. 3 identical versions of each of the ML algorithms will be trained with 3 different training and test data sets. These datasets will each consist of a different type of embedding from those provided in the dataset explained in IV-A2. Concerning the ML algorithms used to define the experiments, we have opted for common models employed on classification problems. Most of the algorithms selected are defined in the PyCaret library [28]

of the Python programming language. More precisely, the following ones:

- Linear Discriminant Analysis (LDA)
- Ridge Classifier (RC)
- MultiLayer Perceptron (MLP) Classifier
- Random Forest Classifier (RF)
- Extra Trees Classifier (ETree)
- Logistic Regression (LR)
- Support Vector Machine - Linear Kernel (SVM-LK)
- K Neighbors Classifier (KNN)
- Decision Tree Classifier (DT)
- Naive Bayes
- Quadratic Discriminant Analysis (QDA)

All of them with the default PyCaret or scikit-learn default configuration, since PyCaret extract some of these algorithms from scikit-learn library. To address the issue of an imbalanced dataset, we utilised the Synthetic Minority Over-sampling Technique (SMOTE), which is integrated within the PyCaret library. This technique generates synthetic samples for the minority classes to balance the dataset.

V. RESULTS

This section will show the experimental results when comparing the embedding representations and the classification algorithms. Additionally, we will show a prototype of our best combination to be used publicly by ethnomusicologic experts to help them with the genre classification problem.

A. EXPERIMENT RESULTS

In this study, we present the findings of two experiments conducted to investigate the effectiveness of the machine learning algorithms exposed in Section IV-B. To reflect the impact of data representation on the performance of classification algorithms, we explored three distinct embeddings for raw audio: Mule, Jukemir, and MERT. Additionally, our experiments involved the evaluation of several widely used machine learning algorithms across these different encodings to test how the embeddings can influence the success of the classification tasks. To evaluate their performance, we used common evaluation metrics in a multi class classification task, such as accuracy, recall, precision, F1 score, Cohen's kappa index, and Matthews Correlation Coefficient (MCC).

The results of these experiments are shown in Table 2. For each embedding, we grouped the ML algorithms applied. Each row shows the evaluation metrics in order. The best algorithms and encoding procedures according to the metrics are highlighted.

In the case of Jukemir embeddings and based on the metrics provided, "Linear Discriminant Analysis" appears to be the best-performing algorithm, as it achieves the highest accuracy, recall, precision, F1-score, Kappa, and MCC. With Mule encodings, the best classifier is the "Extra Tree classifier", as it achieves the best values for all the metrics. Finally, based on the metrics of the algorithms with MERT embeddings, "MLP Classifier" appears to be a strong

performer in terms of accuracy, recall, precision, F1-score, and MCC; however, "Linear Discriminant Analysis" has the highest Kappa index, indicating good discrimination and agreement metrics; additionally, "Ridge Classifier" is the fastest but sacrifice some performance metrics.

With each embedding model, the success of the classification algorithm varies greatly, which means that the latent audio representations extracted with the proposed models (Jukemir, Mule, and MERT) influence the classification process. The combination of Jukemir with the LDA algorithm retrieves the best results according to our metrics. In addition, in the Table 2 is showed that the metrics are higher where Jukemir embeddings are used. Therefore, in general lines, we can affirm that Jukemir embeddings are the best representation for this task since is the one that retrieves the best results.

Comparing the results with previous work is difficult and somewhat unfair, as this is a new dataset. However, it is possible to use work such as [30] as a proxy, which achieved results between 75-80% for a smaller number of classes. Similarly, it would be possible to use datasets commonly used in the literature as a benchmark to evaluate the music genre classification problem, such as GTZAN, where it can be observed that the evaluation metrics obtained are high for the current task. The accuracy of the state of the art in other datasets such as GTZAN in the MARBLE benchmark is around 77.9 for this task.

However, the values are not excellent. This might be caused by two factors. First, we have to consider that the Jukemir, MERT, and Mule approaches exploit advances in latent representations that take polyphonic raw audio as input data. However, in this work, the scores have had to be synthesized to obtain the audio and therefore the characteristics that the latest state-of-the-art models find in audio generated by a performance are not exploited. Our MIDI-to-audio transformation system is not able to fully capture some of the music features, particularly the timbres of traditional instruments like the fiddle or the tin whistle, or the resonances and harmonics. By using a single synthesized instrument, the system can only identify patterns related to the structural elements of the music. However, for musicologists these features are usually enough to identify the genres considered in the paper, as Irish folk dance music always follows very specific features regarding rhythm, phrase structure, tempo and harmony.

In this sense, the differences of the classes in the input data might not be high compared with similar problems in which very different genres (such as pop, rock, heavy metal, etc.) are considered.

To illustrate this problem, we present a visualization of the tunes' representation in embedding form which gave the best results according to our previous analysis in Figure 3. The Figure shows the dataset encoded as a Jukemir embedding plotted using the Linear Discriminant Analysis (LDA) technique with dimension two. One can observe that the different embeddings are mixed; this may

TABLE 2. Results of trained evaluated classifiers.

Embedding	Model	Acc.	Recall	Prec.	F1	Kappa	MCC
Jukemir	LDA	0.8597	0.9824	0.8805	0.8676	0.8239	0.8248
	RD	0.8449	0.8449	0.8678	0.8534	0.8061	0.8071
	MLP	0.8161	0.8161	0.8494	0.8243	0.7692	0.7721
	RF	0.7800	0.7800	0.7836	0.7787	0.7197	0.7200
	ETree	0.7722	0.7722	0.7470	0.7466	0.7053	0.7080
	LR	0.7011	0.7011	0.7995	0.7350	0.6384	0.6440
	SVM-LK	0.6265	0.6265	0.8123	0.6523	0.5598	0.5803
	KNN	0.5384	0.5384	0.6926	0.5688	0.4646	0.4792
	DT	0.4974	0.4974	0.5723	0.5250	0.3923	0.3959
	Naive-Bayes	0.4505	0.4505	0.6094	0.4862	0.3586	0.3722
Mule	QDA	0.4080	0.4080	0.6700	0.2800	0.0840	0.1989
	LDA	0.6423	0.6423	0.7517	0.6809	0.5660	0.5726
	RD	0.0624	0.0624	0.4046	0.0530	0.0312	0.0529
	MLP	0.4853	0.4853	0.6306	0.5038	0.3916	0.4078
	RF	0.7061	0.7061	0.7056	0.7043	0.6260	0.6262
	ETree	0.7280	0.7280	0.7253	0.7231	0.6507	0.6512
	LR	0.0255	0.0255	0.3484	0.0243	0.0085	0.0207
	SVM-LK	0.1144	0.1144	0.1836	0.0759	0.0209	0.0409
	KNN	0.5308	0.5308	0.6613	0.5535	0.4514	0.4656
	DT	0.4878	0.4878	0.5368	0.5053	0.3764	0.3790
MERT	Naive-Bayes	0.5366	0.0492	0.1881	0.0461	0.0172	0.0270
	QDA	0.3959	0.3959	0.5739	0.3203	0.1366	0.1498
	LDA	0.7636	0.7636	0.8203	0.7840	0.7088	0.7117
	RD	0.7187	0.7187	0.7962	0.7439	0.6585	0.6636
	MLP	0.7731	0.7731	0.8062	0.7835	0.7174	0.7195
	RF	0.6498	0.6498	0.6502	0.6459	0.5521	0.5527
	ETree	0.6575	0.6575	0.6577	0.6510	0.5559	0.5570
	LR	0.7442	0.7442	0.8082	0.7669	0.6868	0.6904
	SVM-LK	0.6440	0.6440	0.7979	0.6678	0.5747	0.5981
	KNN	0.4584	0.4584	0.6160	0.4775	0.3794	0.3951
	DT	0.3508	0.3508	0.4187	0.3723	0.2265	0.2299
	Naive-Bayes	0.2435	0.2435	0.4305	0.2733	0.1606	0.1720
	QDA	0.5839	0.5839	0.7322	0.5030	0.3897	0.4442

be due to two reasons: (I) the audios from which these embeddings have been obtained are synthesized audios, and (II) the classes that they represent are subgenres of traditional Irish music; moreover, most of them are dance music, which is very similar to each other in terms of rhythmic patterns, musical structure, instrumentation, and harmony. Therefore, we presume the high similarity among the data has affected the results obtained in the previous experiment.

Moreover, Figure 4 shows the confusion matrix for the previous combination. We can observe that the genres air, barndance, clog, fling, march, and mazurka yielded the poorest classification results. We attribute this primarily to the limited number of available tunes (fewer than 3000), which may prevent the system from fully capturing the distinctive features of these genres. Below, we provide a musicological analysis to rationalize the observed confusions in the classification results.

The system often misclassifies airs as jigs or reels. This might occur because airs encompass a wide variety of structures, meters, and melodies, making them highly diverse. Some airs are derived from dances or laments and can resemble jigs and reels in terms of meter, instrumentation, and musical form.

Barndances, which are in 4/4 time like reels and binary like marches, are frequently confused with these genres.

Their typically fast tempo could also contribute to the misclassification of reels and marches as barndances.

Confusion between clogs and barndances is also noted. Both genres share a “rural” character, influencing their structure and potential rhythmic patterns, leading to this misclassification.

The genre of highland fling is completely mistaken for strathspeys and waltzes. Despite poor classification results, the confusion with strathspeys is noteworthy. Both genres originate from the Scottish Highlands, share a 4/4 meter, and include characteristic dotted rhythms. Further analysis with a larger dataset could elucidate the distinct features between Scottish and Irish traditional music.

Mazurkas are commonly confused with waltzes, as both are 3/4 dances with similar structures. The key difference is that mazurkas emphasize the second beat, whereas waltzes accentuate the first beat of the three-beat measure.

Finally, while the classification of slides is generally accurate, there is notable confusion with jigs. This is understandable since slides are sometimes considered minor variations of jigs, differentiated mainly by their faster tempo and alternation of long and short rhythmic figures.

This detailed analysis underscores the need for a larger dataset to improve genre classification accuracy in some cases and further explore the nuanced characteristics of traditional music genres.



FIGURE 3. Jukemir dataset's LDA representation.

		LinearDiscriminantAnalysis Confusion Matrix														
True Class	air	49%	0%	0%	0%	2%	20%	0%	0%	0%	15%	0%	5%	0%	0%	10%
	barndance	0%	62%	0%	0%	0%	0%	15%	0%	4%	16%	0%	0%	1%	0%	1%
	clog	0%	23%	54%	0%	0%	0%	8%	0%	0%	0%	0%	8%	8%	0%	0%
	highland fling	0%	0%	0%	0%	0%	0%	0%	0%	20%	0%	0%	0%	60%	0%	20%
	hornpipe	0%	2%	0%	0%	93%	0%	1%	0%	0%	3%	0%	0%	0%	0%	0%
	jig	0%	0%	0%	0%	0%	92%	0%	0%	0%	4%	2%	0%	0%	0%	1%
	march	0%	15%	0%	0%	0%	1%	59%	0%	8%	8%	0%	0%	5%	0%	3%
	mazurka	0%	0%	0%	0%	0%	1%	66%	0%	0%	0%	1%	0%	0%	0%	32%
	polka	0%	4%	0%	0%	0%	0%	2%	0%	89%	3%	0%	0%	1%	0%	1%
	reel	0%	6%	0%	0%	0%	0%	3%	0%	1%	87%	0%	0%	1%	0%	0%
	slide	0%	0%	0%	0%	0%	22%	0%	0%	0%	0%	75%	2%	1%	0%	1%
	slip jig	0%	0%	0%	0%	0%	5%	0%	0%	0%	1%	3%	88%	0%	1%	1%
	strathspey	0%	6%	0%	0%	0%	0%	8%	1%	1%	11%	0%	0%	72%	0%	1%
	three-two	0%	0%	0%	0%	0%	0%	0%	3%	0%	0%	1%	0%	0%	85%	11%
	waltz	0%	1%	0%	0%	0%	3%	1%	7%	0%	0%	0%	0%	0%	1%	85%
		air	barndance	clog	highland fling	hornpipe	jig	march	mazurka	polka	reel	slide	slip jig	strathspey	three-two	waltz
		Predicted Class														

FIGURE 4. Confusion matrix for Jukemir-LDA classifier.

Despite all, utilizing real polyphonic audio recordings of Irish folk music would probably enhance the accuracy of our results. This improvement is attributable to our most effective model, Jukebox, which achieves optimal performance when trained on raw, authentic audio. Such training preserves the texture and variety inherent in Irish folk music.

B. TRADITIONAL IRISH ARCHIVE TOOL

In order to encourage public participation, give direct use to our research, and, at the same time, increase our dataset, we have created a tool that can be accessed.³

³<https://zenodo.org/doi/10.5281/zenodo.10659379>

It consists of an interface where a user can upload a score and classify it. The input is a file formatted in MusicXML, ABC notation, or MIDI. We will automatically convert the input into ABC notation and apply our classification approach. The output will be the genre of the tune uploaded according to our algorithm and its accuracy rate.

The tool is implemented in Python and accessible from Google Colab Notebook. The embedding used is Mert, as the Jukemir embeddings require too much RAM and VRAM to run on a free tier of Google Colab. The algorithm included is the combination that worked best according to our tests shown in the previous section.

VI. CONCLUSION

In this study, we have successfully constructed a tool harnessing the capabilities of advanced classification algorithms to discern subgenres within traditional Irish music environments using scores instead of raw audio. By leveraging state-of-the-art data mining techniques, we have tested three different embeddings to analyze their accuracy in the classification problem with this kind of input data (XML/ABC notation).

Our work drew upon a diverse dataset extracted from two reputable repositories, TheSession and ITMA, ensuring the inclusion of a wide spectrum of Irish traditional music recordings, above all related to dances, which constitute a main genre within this tradition. To effectively process and analyze this eclectic collection, we applied three distinct encoding methods: Jukemir, Mule, and MERT. These encoding techniques facilitated the transformation of raw musical data into a latent representation for computational analysis, allowing us to extract associations that reveal possible groups among the data.

To comprehensively assess the performance of genre classification algorithms, we designed and executed an experiment that involved the comparative evaluation of seven prominent machine learning algorithms. Moreover, we explored the impact of the three embedding techniques on the classification metrics, showing the effectiveness of each method in classifying musical notated files in different subgenres.

Despite some limitations of our study, namely the use of music notation data rather than real audio recordings, and the representation of only a subset of possible genres within Irish traditional music, we highlight our contributions to the research community. We have cultivated a novel dataset encoded with cutting-edge techniques and have also demonstrated the validity of these methods in accurately classifying Irish traditional music. Our findings underscore the potential for further advancements in the realm of music genre classification and provide a solid foundation for future exploration and inquiry within this musical domain.

As a future work, we aim to explore the possibilities of fusing multiple data modalities, including both raw audio and symbolic notation data, to create a more robust and comprehensive dataset. Compared to synthesized raw audio, symbolic notation preserves more characteristics such as time signature, tempo, key signature, mode, timbre difference between various instruments, etc. This approach could provide a more holistic representation of Irish traditional music and further improve classification results. We consider that symbolic notation of the data can better preserve the distinctive features of each of the sub-genres within traditional Irish music. Additionally, we aim to develop hierarchical genre classification models that account for the inherent subgenres and variations within Irish traditional music. This approach would enable a finer-grained classification and better capture the diversity of musical styles.

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TREASA HARKIN is a member of Irish Traditional Music Archive (ITMA). Her professional experience covers librarianship, archiving, audiovisual technology, IT, broadcasting, education, design, management and business, and various European languages.

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